

The Roles This Book Trains — Aasan Summary (Roman Urdu + English)

Source: *The AI Agent Factory* — “The Roles This Book Trains”
(Panaversity)

🌀 Bunyadi Idea

Yuval Noah Harari ka sawal: “**Agar hume nahi pata job market 10 saal baad kaisi hogi, to aaj hum kya sikhayein?**”

Purana jawab tha: “**coding sikho.**” Yeh jawab tootrha hai — AI khud coding seekh raha hai, ho sakta hai jald hi zyada tar code AI hi likhe.

Is book ka jawab: Syntax-writers mat bano. Woh log bano jo **machine ke upar khare hon** — jo decide karein kya banana hai, AI Workers ko supervise karein, aur jo wapas aaye usay verify karein. **Judgment, specification, aur deployment** — yeh cheezein machine absorb nahi kar sakti.

📊 3 Levels, 1 Line

Aap (insaan) → AI Worker (jo aap banate ho) → Poori Company (jo un Workers se milkar banti hai)

Market ne is poori line ko carry karne wale insaan ka naya naam diya hai: **Forward Deployed Engineer (FDE)**.

Zaroori baat: “FDE” batata hai aap **kahan** kaam kar rahe ho, **kya jaante ho** yeh nahi. Client ki company ke andar kaam karo — market aapko FDE kehta hai. Wahi company aapko hire kar le — same skills aapko unka **AI-Native Company Architect** bana deti hain.

🏠 Baseline: Har Koi Yahan Se Shuru Karta Hai

Foundations — browser tab mein, koi installation nahi: prompt karna, do document languages, skills/connectors, AI era mein sochna.

Mode 1 Practitioner — koi title nahi, sirf ek proficiency. General agent se apna hi kaam tez karna.

Mode 2 — asli job titles yahan hain: AI Workers **manufacture** karna jo aapke liye kaam karein.

🔗 4 Core Roles — Ek Pipeline

Outcome Architect (kya) → Digital FTE Builder (banao) → AI-Native Company Architect (system) → Cloud AI Engineer (chalao)

Apni company ke andar chalao to yeh 4 roles hain. Client ki company ke andar, ek insaan carry kare, to woh **FDE** ban jata hai.

Role	Kya Karta Hai
Outcome Architect	Intent ka malik. Kya banana hai aur kyun , spec likhta hai, “correct” ka matlab set karta hai
Digital FTE Builder	Poora unit banata hai — spec, SKILL.md, agent architecture, MCP, evaluation
AI-Native Company Architect	Poori company design karta hai — Two-Layer Model, workforce, nervous system
Cloud AI Engineer	Workers ko production mein chalata hai — Azure, Kubernetes, scale

⚠ **Zaroori Shift: Intent Bottleneck**

Pehle: **1 Product Manager per 8 engineers**. Ab AI coding agents se engineer ka output 5x barh gaya — lekin decide karne wale insaan same reh gaye. Isliye **1 insaan ab 20 engineers ka kaam feed karta hai!**

Building scale ho gayi. Decide karna scale nahi hua. Isi liye **Outcome Architect** company ki sabse important seat ban raha hai.

 Forward Deployed Engineer (FDE) — Woh Role Jo Market Dhoond Nahi Pa Raha

Palantir (bara data analytics company) ne yeh role 2010s mein banaya — pehle “Deltas” bulate the. FDE **customer ki company ke andar** jakar, unke real data par, real problem solve karta hai — jaise doctor jo aapke chart ko doosre shehar se nahi, balke **aapke saamne kamre mein baith kar** dekhta hai.

Palantir ka farq: Normal developer = “ek capability, kai customers” (ek feature sab ko do). FDE = “ek customer, kai capabilities” (ek client ke sath jo bhi chahiye).

Kevin Bai Ka 2×2 Test — Kya Aapko FDE Chahiye?

	Technical Buyer	Non-Technical Buyer
Configurable Product	Kuch khaas nahi chahiye	Sales-led motion (Jira, Slack)
Technical Product	Docs + dev relations (GitHub)	★ FDE Chahiye! (Palantir ka corner)

Agentic turn: Ab har platform agentic hai, isliye customizable hai — matlab har vendor is 4th cell mein aa raha hai. Isi liye 2026 mein demand explode ho gayi!

Numbers Jo Loud Hain

- FDE job postings **800%+** barh gayin (9 mahinon mein), candidates sirf **~50%**
- 2026 mein ek hafte mein AWS (\$1B) + Microsoft (\$2.5B) = **\$3.5 billion** naye FDE units mein
- OpenAI ne **\$4 billion** ki alag deployment company banai
- **95% enterprise AI pilots** koi measurable return nahi dete (MIT study) — lekin **outside partners** ke sath kaam karne wale 67% deployment tak pahunchte hain, akele karne walon mein sirf 33%

Pay Bands (Real Numbers)

Level	Pay
Median	~\$190,000/saal
Anthropic postings	\$200K-\$300K
Senior/Staff	\$450K-\$600K

🔒 Do Band Darwaze

1. **Lock-in (Client ko Nuksaan)** — Har vendor ka FDE apne platform mein sab jorta hai. Client permanently phans jata hai — jaise plumber jo sirf ek brand ki pipes lagata hai.
2. **Hiring Door (Aapko Nuksaan)** — Vendor FDE senior door hai (5+ saal, US/Europe address). Duniya ke zyada tar developers bahar rah jate hain.

Andrew Ng: Clients ko aisa FDE dhoondna mushkil hai jo kisi ek vendor se bandha na ho — kyunki lock-in hi vendor ke liye role ka **poora point** hai!

🔑 Hamara Jawab: Vendor-Neutral Vertical FDE

2 Systems of Record leke jate ho: 1. **Method** — poori book, sab ko multi hai (aapne nahi banayi) 2. **Profession** — ek vertical, ek jurisdiction, aapne khud expert ke sath banayi

Dono MCP bolte hain, isliye aapka agent dono ek sath padh sakta hai — **koi integration ka jhagra nahi.**

“Dev Shop” Ka Aitraaz — Bai Ka Sabse Sakht Sawal

Bai kehta hai: agar har FDE scratch se banaye, to woh **FDE function nahi, dev shop hai** — maintenance cost profit kha jayegi. Uska sawal: **“Kya mere paas platform hai, ya main ek banane ke liye tayyar hoon?”**

Jawab: Vendor-neutral FDE ke paas bhi shared primitives hain — bas woh **code nahi**, balke **2 governed Systems of Record** hain! Method har client par same hai (like a platform). Profession ek vertical mein har client ke sath **motti** hoti jati hai.

👥 Pod of One

AWS ek client ko **5-6 engineers ki team** bhejta hai, ~45 din ke liye. Yeh book **team ki jagah 1 insaan** banati hai — kyunki baaki team members ab **Digital FTEs (AI Workers)** hain jo yeh insaan banata aur chalata hai!

⚠ **Risk:** Single point of failure. Agar aap 2 hafte ke liye unreachable ho, kya koi aur graduate aapke repository se aapka engagement sambhaal sakta hai? Agar nahi, to yeh “pod of one” nahi, “bus factor of one” hai — sab kuch **artifacts mein likha hona chahiye** (spec, evals, corpus, runbook).

📦 Teesra Darwaza: Freelance FDE

Upwork par **FDE category exist karta hai**, lekin uske andar supply generalists hai — kisi ne bhi apne aap ko “FDE” ki tarah describe nahi kiya! **Yeh khaali shelf hi opportunity hai.**

Freelance Bands: Pehla integration \$2K-\$5K, custom \$5K-\$15K, enterprise \$15K+, retainer \$4K-\$10K/mahina, consulting \$150-\$250/ghanta.

3 khaas cheezein is darwaze ki: 1. Yeh **vendor-neutral FDE ka apna market hai** (vendor FDE freelance nahi kar sakta) 2. Retainer = Digital FTE subscription model, outside se chalayi hui 3. **Koi border nahi** — Karachi, Lagos, Bangalore se equal kaam mil sakta hai

Direct-Hire Freedom: Client aapko directly hire kar le, to “FDE” title khatam ho jati hai — kaam wahi rehta hai, sirf address change hota hai. Aap unke **AI-Native Company Architect** ban jate ho. **Sirf vendor-neutral FDE yeh trip kar sakta hai** — vendor ka FDE company chhorte hi platform peeche reh jata hai.

👤📦 Subject Matter Expert as Skill Author

Market ne is role ko naam nahi diya abhi tak. Accountant, lawyer, ya supply-chain expert jo apna **tacit judgment SKILL.md** mein likh de aur Digital FTE ka “knowledge engine” ban jaye.

Kaam: Woh rule jo aap bina soche follow karte ho — jaise auditor kaise decide karta hai konsi transactions flag karni hain — usay itni precisely likho ke agent execute kar sake, phir agent ke jawabbaat ko apne se match karo, jab tak match na ho jaye.

Real Example (Yousuf Imran): Google mein \$986,000/saal kamane wala salesperson, apna 20-saal ka sales judgment AI product mein encode karne ke liye company chhor deta hai — kyunki **equity uske paas honi chahiye, kisi aur ke paas nahi.**

Expert aur Engineer ek dusre ke bina adhuraray hain: Expert 20 saal ka license laata hai, Engineer method aur build laata hai. Yehi “vertical” ka bunyad hai.

🔌 Connector aur Plugin Engineer

Market isay 5 alag naamon se pukarti hai (MCP engineer, integrations engineer, wagera) — **asal mein ek hi kaam hai, do addresses par:**

- **Connector-native app** → chat app (claude.ai) ko extend karta hai

- end users ke liye
- **Plugin** → coding agent (Claude Code, OpenCode) ko extend karta hai — builders ke liye

Dono ka core artifact same hai: **MCP server**.

🛡️ 3 Supporting Roles

Role	Kaam
Evals Engineer	AI Worker ko production se pehle “crash-test” karta hai — jaise gari ka crash test
AI Governance Officer	Rules likhta hai: AI khud kya decide kar sakta hai, kab insaan ki approval chahiye, kya kabhi touch nahi karna
Digital FTE Supervisor	Human-in-the-loop — reviewer jo kaam check karta hai, audit trail jiska naam hai

📖 Book Jahan Rukti Hai

Role	Kya Trains Hoti Hai	Kya Nahi
LLMOps Engineer	Fine-tuning (last resort ki tarah)	Foundation model banana (from scratch)
Harness Engineer	Existing runtime use karna (OpenAI SDK, Claude managed agents)	Naya runtime banana
AI Data Engineer	Agent-facing layer (Postgres, pgvector, MCP)	General data pipeline/warehouse engineering

🧠 Doosra Axis: Aapka “Type”, Sirf Seat Nahi

Boris Cherny (Claude Code ke banane wale) ne 5 archetypes diye — koi bhi job title nahi:

Archetype	Kaam	Roughly Rhymes With
Prototyper	Nayi ideas banata hai, zyada tar ship nahi hote	Outcome Architect
Builder	Prototype ko production-grade banata hai	Digital FTE Builder
Sweeper	System simplify karta hai, cleanup, optimize	Evals Engineer
Grower	Product ko market-fit ki taraf le jata hai	—

Maintainer

Mature system ko
secure/reliable
rakhta hai

Cloud AI Engineer,
Supervisor

Zyada tar log 2-3 archetypes mein span karte hain, ek nahi. Yeh roles batatay hain “kaam kahan hai,” aapke archetypes batate hain “aap konsi seat le sakte ho.”

📖 Appendix A — FDE Résumé Ke 6 Signals

Delivery signals: 1. Shipped production systems (sirf features nahi)
2. Quantifiable impact (numbers ke sath) 3. Direct customer exposure

How-you-deliver signals: 4. Messy-data work (real client environments kabhi clean nahi hoti) 5. Ownership under ambiguity 6. AI/LLM depth (RAG, agents, evals)

3 Rewrites: - “Built ETL pipeline” → “Shipped a pipeline that cut client’s close from 5 days to 2” - “We” nahi, **“I”** likho - Competitive-programming awards hatao — portfolio dikhao (deployed Worker, plugin, connector app)

🔧 Appendix B — FDE Interview: Loop Aur Jaal

5-8 stages, 3-6 hafte: recruiter screen, hiring-manager screen, coding round, system-design, **decomposition case study**, **client simulation**, behavioral round.

Decomposition Round — Sabse Bara Filter

Vague problem milta hai (jaise “shehar ka emergency response time kam karo”). **Sabse zyada rejection:** seedha solution dena (“main XGBoost use karunga”). **Sahi tareeqa:** goal clarify karo, stakeholders naam lo, data map karo, risk se sequence karo, sabse patla end-to-end skeleton propose karo.

Client Simulation — Doosra Filter

Interviewer customer ban jata hai (frustrated ya non-technical). Aapko bad news deni hai, governance-compromising request se inkar karna hai, business language mein baat karni hai.

5 Red Flags: Bina clarify kiye solve karna, cost/constraints ignore karna, thin deployment story, compliance vocabulary ka na hona, customer instinct ki kami.

★ Sabse Zaroori Baat

Yuh book ke saare roles isi ek pattern se bantay hain: Agentic era kaam ko **kai roles mein baant raha hai** — Workers banana, chalana, governing karna, judgment sikhana. **Aapke liye zaroori sawal:** Aap kahan khare ho, aap kaunse archetypes mein span karte ho, aur yeh book aapko wahan se kitna aage le ja sakti hai?

Quick Glossary

Term	Aasan Matlab
FDE	Client ki company ke andar embed hone wala engineer, jab tak outcome prove na ho
Outcome Architect	Intent ka malik — kya banana hai aur kyun
Digital FTE Builder	AI Worker banane wala
AI-Native Company Architect	Poori company design karne wala
Cloud AI Engineer	Production mein chalane wala
Skill Author	Apna domain judgment SKILL.md mein encode karne wala expert
Pod of One	Ek insaan + kai AI Workers, jahan pehle 5-6 log kaam karte the
Bus Factor of One	Risk jab sirf ek insaan ke sar mein saari knowledge ho
Archetype	Cherny ke 5 kaam ke tareeqay (Prototyper, Builder, Sweeper, Grower, Maintainer) — job title nahi

Yeh summary poore “The Roles This Book Trains” page ka overview hai — Loop, Harness, Ecosystem, aur How to Get Paid ke baad, “main is sab mein se kaun hoon” ka jawab.